



NATIONAL DEVELOPMENT COMPLEX

Deputy Directorate General of QMS

High Energy Radiation Laboratory, 3E

TEST REPORT

Test Report No: TR (039)/HERL/23

Job Card No: JC (052)/HERL/23

Date: January 27, 2023

INDENTER'S DATA			
Name:	Muhammad Shafique	Designation:	SRA
Group & Site:	Winding / 3C	Directorate:	ADP
Organization:	NDC	Contact No:	3641, 3708

GENERAL DATA			
Sample Name:	Nose (S1AM)	ID No:	RD-19C (a)
Sample receive Date:	January 23, 2023	Test Date:	January 25, 2023
Geometrical Shape:	T Shape	Quantity:	01
		Material:	3D Carbon
		Thickness:	147 mm

TEST DATA			
Test Type: General	Test Name: Computed Radiography Computed Radiography (After Machining)	Test Method: ASME V	Technique: SWSI
Radiation Source:	X-Ray Tube Smart 160 E	Focal Spot:	1.5 mm
Qualification Standard:	NDC-CPS/Comp/QM/3DCarbon/CCC-01 (Rev:01)	SFD:	100 cm
		Temperature:	24.6 °C
		Humidity:	42 %
		IQI:	GOST 7512 (CP-04)
		Sensitivity:	2%

Image Acquisition Parameters:

- Detector Type: Carestream HR Imaging Plate
- Processing Equipment: Carestream HPX-1 CR System
- Imaging Software: Carestream Industrex Digital Viewing Software (Version 4.1)

Test Purpose: To check internal defects

Results:

- No internal defects have been observed in the Nose.
- Scattered surface pitting/ porosity of size 1.0 mm and depth < 0.5 mm has been observed on the nose.

Status: OK

Remarks:

- ID # is written with marker on the 3D Carbon nose.
- 3D Carbon Rod with ID # RD-19C has already been radiographed. [Ref: TR (088)/HERL/22; Dated: 14-03-2022]. It was cut into parts and RD-19C (a) is its sub part.
- Pitting depth of nose tip was measured by Metrology. [Ref: TR (040)/PML/23, dated: 18-01-23].
- These results relate only to the item mentioned and tested above.
- This test report shall not be reproduced except in full, without written approval of QC Directorate.

Test Performed by: Mr. Abdul Wadud. (STO)

M. Zaheer Abbas. HOD (CT)

Verified by: Mr. Gohar Farooq. HOD (RT)

Contact No. 3268 & 3269

Reviewed By:

PD (NDT):

Mr. Altaf Hussain.

Approved By:

Dir. (NDT):

Syed Jaber Hussain Shah

HERL/FM/02, Issue 09 (03-04-17)

(End of report)

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Prepared By: GM Reviewed & Recommended by: GM Approved By: GMR/ DCM (NDT)